

Rebuilding Delivery Performance at Apex Manufacturing

A specialty vehicle upfitter where nothing was shipping on time, rebuilt in fifty days around three changes: make the work visible, move the quality check into the department that does it, and resequence the floor.

ORGANISATION	ROLE	SECTOR	PERIOD
Apex Manufacturing	Managing Director	Specialty vehicle upfitting	2023, first 90 days

The situation

Apex Manufacturing built specialty vehicles to order on a single upfitting line. The order book was healthy and the work was good, but almost nothing left the building on the date it had been promised. The board wanted to know whether the problem was the people, the pricing or the plant.

It was none of those. The company had no way of knowing where any job was. Status lived in the heads of the people doing the work, so a job was discovered to be late at the point it became late, which is the point at which nothing can be done about it. Quality was inspected once, at the end of the line, so a wiring error made in the second hour of a four-day build was found on day four and corrected by people who had not made it. Assembly lost one to two hours every day to rework caused upstream. Twenty-five people worked a floor laid out for a product it no longer built, and roughly \$3M of annual spend was committed against forecasts nobody trusted.

What was done, in order

- Week one: walked the line and timed every stage before changing anything. The bottleneck was not where the team believed it was.
- Week two: built the company's first WIP report. Every active job, its stage and its committed date on one page, updated daily. Bottlenecks that had taken weeks to surface became visible within hours.
- Weeks three to five: replaced end-of-line inspection with departmental QC standards. Written pass criteria per department, and work did not advance until the department that did it signed it off.
- Weeks four to seven: resequenced the floor into a single direction of travel, with manufacturing templates for the repeating build types, cutting six queues back to one line.
- Throughout: shifted material purchasing from overstock ordering to deposit-based buying against confirmed orders, taking carrying cost down without risking a stockout.

Results

100% on-time delivery, reached within 50 days	25 to 13 headcount, with output and delivery both up	0 hrs daily assembly rework, from 1 to 2 hours	\$2M-\$3M new pipeline converted in the same quarter
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MEASURE	ON ARRIVAL	DAY 50	HELD AT
On-time delivery	41%	100%	100%
Open WIP items	64	19	17 to 22
Time to see a bottleneck	2 to 3 weeks	Same day	Same day
Daily assembly rework	1 to 2 hours	None	None
Headcount	25	13	13
Material carrying cost	Overstock	Deposit-based	Deposit-based

Why the headcount fell

The reduction from twenty-five to thirteen was an outcome, not a target, and the order of events matters. Nobody was removed to hit a number. Once the rework stopped, once jobs stopped being built twice, and once the floor ran in one direction, the line needed fewer hands to produce more than it had before. The roles that disappeared existed to absorb the disorder.

What made it hold

Every one of these changes has been made at other plants and quietly reversed within a year, usually because the standard belonged to the person who introduced it. Here the QC standards were written by the departments that had to meet them, and supervisors were taught the reasoning behind each criterion rather than told which correction to apply. Four of the seven department leads at the end of the period had been operators at the start of it.

"We brought Daniel in because nothing was shipping on time. Within fifty days every active account was delivering on schedule, and he did it while reducing the team from twenty-five to thirteen. He does not just fix the number, he rebuilds the system underneath it so it holds."

Ellen Brightwater, Board Member, Apex Manufacturing